

Dataset Sources and Access

- **World Happiness Report - 2024**
 - Source: [World Happiness Report - 2024](#)
 - Coverage: Not Available
 - Licensing/Terms: Publicly available through the Kaggle website; [CC0 Public Domain](#)
- **Global Country Information Dataset 2023**
 - Source: [Global Country Information Dataset](#)
 - Coverage: Not Available
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Data Cleaning and Integration

- Removed unrecognized territories/countries and countries with missing/irrelevant values:
 - Included Taiwan (The only exception, because they had the important values and it is a wealthy country that can help us with our research)
 - Cleaned out universally unrecognized countries like Palestine
 - Cleaned out countries that were missing values
- Removed countries that were missing important variables like GDP, Minimum Wage, etc.
- Renamed certain countries to match the name on the other dataset. For example, there are multiple names for the USA.
- Did not need to reduce the dataset size since it fits within GitHub, fortunately.

Selected Attributes for Integrated Dataset

1. **Record ID:** Unique identifier for each observation.
2. **Country name:** Name of the country.
3. **Abbreviation:** Abbreviation of country
4. **Regional indicator:** Region to which the country belongs.
5. **Capital/Major City:** Name of the capital or major city.
6. **Population:** Total population of the country.
7. **Density (P/Km2):** Population density measured in persons per square kilometer.
8. **Land Area (Km2):** Total land area of the country in square kilometers.
9. **GDP:** Gross Domestic Product, the total value of goods and services produced in the country.
10. **Minimum Wage:** Minimum wage level in local currency.

11. **Currency_Code**: Currency code used in the country.
12. **Happiness score**: The happiness score for each country, based on responses to the Cantril Ladder question that asks respondents to think of a ladder, with the best possible life for them being a 10, and the worst possible life being a 0.
13. **Log GDP per capita**: The natural logarithm of the country's GDP per capita, adjusted for purchasing power parity (PPP) to account for differences in the cost of living between countries.
14. **Social support**: The national average of binary responses(either 0 or 1 representing No/Yes) to the question about having relatives or friends to count on in times of trouble.
15. **Healthy life expectancy**: The average number of years a newborn infant would live in good health, based on mortality rates and life expectancy at different ages.
16. **Freedom to make life choices**: The national average of responses to the question about satisfaction with freedom to choose what to do with one's life.
17. **Generosity**: The residual of regressing the national average of responses to the question about donating money to charity on GDP per capita.
18. **Perceptions of corruption**: The national average of survey responses to questions about the perceived extent of corruption in the government and businesses.
19. **Dystopia + residual**: Dystopia is an imaginary country with the world's least-happy people, used as a benchmark for comparison. The dystopia + residual score is a combination of the Dystopia score and the unexplained residual for each country, ensuring that the combined score is always positive. Each of these factors contributes to the overall happiness score, but the Dystopia + residual value is a benchmark that ensures no country has a lower score than the hypothetical Dystopia.
20. **Tax Revenue (%)**: Tax revenue as a percentage of GDP.
21. **Unemployment Rate**: Percentage of the labor force that is unemployed.
22. **Life Expectancy**: Average number of years a newborn is expected to live.
23. **Birth Rate**: Number of births per 1,000 population per year.
24. **Fertility Rate**: Average number of children born to a woman during her lifetime.
25. **Infant Mortality**: Number of deaths per 1,000 live births before reaching one year of age.
26. **Latitude**: Latitude coordinate of the country's location.
27. **Longitude**: Longitude coordinate of the country's location.

Data Quality

- Missing GDP values: 0
- Missing Happiness scores: 0
- Missing Minimum wage: 0
- Missing Dystopia values: 0

Storage & Versioning

- Original CSV files stored in GitHub
- Edit histories from OpenRefine stored in a GitHub repository for reproducibility

Licenses for Data & Software

Field / Artifact	Type	Description / Notes	License / Terms of Use
<i>Python</i>	Software	Programming language used for analysis, data cleaning, and visualization	Python Software Foundation License Version 2
<i>NumPy</i>	Software	Library for numerical computation and array operations	BSD-style, free to use, modify, distribute, disclaimers apply
<i>Pandas</i>	Software	Library for data manipulation, analysis, and cleaning	BSD-style, free to use, modify, distribute, and must retain the copyright notice
<i>Matplotlib</i>	Software	Library for creating visualizations and plots	BSD-style, free to use, modify, distribute
<i>World Happiness Report - 2024</i>	Dataset	Contains landmark survey of the state of global happiness	Publicly available; CC0 Public Domain
<i>Global Country</i>	Dataset	Contains information like demographic statistics, economic indicators, environmental factors,	Publicly available; Attribution 4.0 International (CC BY 4.0)

Information
Dataset 2023

healthcare metrics, education
statistics, etc.